
SCITI 2026

Methodology Paper

How we score Thai smart cities — transparently and reproducibly.

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Authors: Nimmanphatcharin, Prathombutr, Siddhichai, A, Treethidtaphat

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1. Three Foundational Concepts

Citizens at the center, technology in the background. Cities are for people. A smart city should not force residents to be on smartphones 12 hours a day. The evidence of success lies in measurable outcomes: less travel time, worry-free healthcare, more spare time.

People-Public-Private-Partnership (PPPP). Not PPP but PPPP. The extra P is for People. Without citizen buy-in, you build infrastructure nobody uses.

Simultaneous physical + digital development. Digital platforms are meaningless if goods and services cannot physically reach people.

2. Two-Five-Seven Framework

Two types: Smart Livable City (existing city + tech) vs Smart New City (greenfield, e.g., EEC).

Five criteria: 1) Vision + boundaries. 2) Infrastructure + investment plan. 3) City Data Platform (CDP) + cybersecurity. 4) Urban systems + service projects. 5) Management model + public participation.

Seven dimensions: Environment (must-have), Economy, Mobility, Energy, People, Living, Governance.

3. Scoring Methodology

The SCITI composite score is a weighted average of 7 pillars, each scored 0-100 from verifiable public data:

Pillar	Weight	Key Signals
Livability	25%	Housing, transit, infrastructure quality
Economy	20%	GPP per capita, income, employment growth
Safety	15%	Crime rate per 100K, disaster resilience
Wellbeing	15%	Hospital beds, education access, health outcomes
Environment	10%	PM2.5 annual average, green coverage, water quality
Hospitality	10%	Cultural vibrancy, community warmth, tourism
Digital	5%	IoT deployment, data platforms, AI adoption

Composite = $\text{Sum}(\text{Pillar}_i \times \text{Weight}_i) / \text{Sum}(\text{Weights})$

Alpha $\geq$ 65 (operational infrastructure). Beta 45-64 (building, partial). Gamma $<$ 45 (planning stage).

4. Data Sources (15+)

- NSO — population, income, employment (Annual)
- NESDC — Gross Provincial Product per capita (Annual)
- PCD Air4Thai — PM2.5 from 70+ stations (Hourly)
- GISTDA — satellite green coverage, urban sprawl (Quarterly)

- World Bank — GDP growth, Gini coefficient (Annual)
- ADB — infrastructure finance, ACGF (Periodic)
- Open-Meteo/Copernicus — climate cross-validation (Daily)
- Royal Thai Police — crime statistics per province (Annual)
- DOPA — population registry, migration (Monthly)
- depa citydata.in.th — IoT sensors, citizen engagement (Real-time)
- MOPH — hospital beds, healthcare access (Annual)
- BOI — FDI approvals, investment zones (Quarterly)
- ONEP — environmental performance index (Annual)
- Royal Forest Dept — forest cover by province (Annual)
- Field observation — direct site verification (Ongoing)

5. Data Confidence

Each city receives an automatic data confidence rating based on metric completeness: **High** (5+ of 6 key metrics filled), **Medium** (3-4), **Low** (<3). Cities with low confidence are flagged — their scores carry higher uncertainty.

6. Financial Toolkit

15 financial instruments are matched to each city based on tier, development stage, and pillar gaps. Sources: ASEAN Smart City Financing Toolkit, ADB ACGF, Thailand SEC Green Bond Framework, BOI S-Curve. Each city also receives a bankability assessment (0-100) covering revenue base, institutional capacity, project pipeline, private interest, and risk profile.

7. Standards Alignment

- UN-Habitat City Prosperity Initiative (CPI)
- ISO 37122:2019 — Sustainable Cities: Smart City Indicators
- UN SDG 11 — Sustainable Cities and Communities
- ASEAN Smart Cities Framework (ASCF) 2018
- ASEAN Smart Cities Action Plan (ASCAP) 2021-2025
- New Urban Agenda 2016
- Thailand PPP Act B.E. 2562 (2019)